cognitive artificial intelligence (AI), machine learning (ML), quantum communication, and advanced swarm coordination to operate independently across multiple domains: air, land, sea, and space. These drones represent the most advanced stage in UAV evolution, capable of self-learning, adaptive decision-making, and mission execution without continuous human control or ground-based command dependency.

Unlike earlier generations that relied on pre-programed algorithms or operator-led missions, fifth-generation drones are equipped with cognitive computing frameworks that enable them to perceive their environment, analyze real-time data, and make strategic choices dynamically. They embody true autonomy, combining AI-driven situational awareness, distributed data networks, and neural flight control systems to perform cooperative tasks with other unmanned or manned platforms.

This generation also integrates quantum-encrypted communication systems for secure, latency-free information exchange and digital twin technologies for predictive maintenance and mission simulation. The use of self-healing materials, hybrid propulsion systems, and energy-efficient architectures allows these drones to achieve long-duration endurance, stealth, and adaptability under extreme operational conditions.

6.1.2 Basic characteristics of fifth generation of drones

The fifth generation of drones embodies the pinnacle of autonomy, intelligence, and interconnectivity in unmanned aerial systems (UAS). These drones are characterized by their ability to operate cognitively, collaboratively, and cross-domain, transforming from remotely controlled systems into self-governing intelligent entities capable of independent reasoning and cooperative mission execution. Their design philosophy integrates advancements in AI, quantum computing, materials science, propulsion, and communication, resulting in systems that redefine the